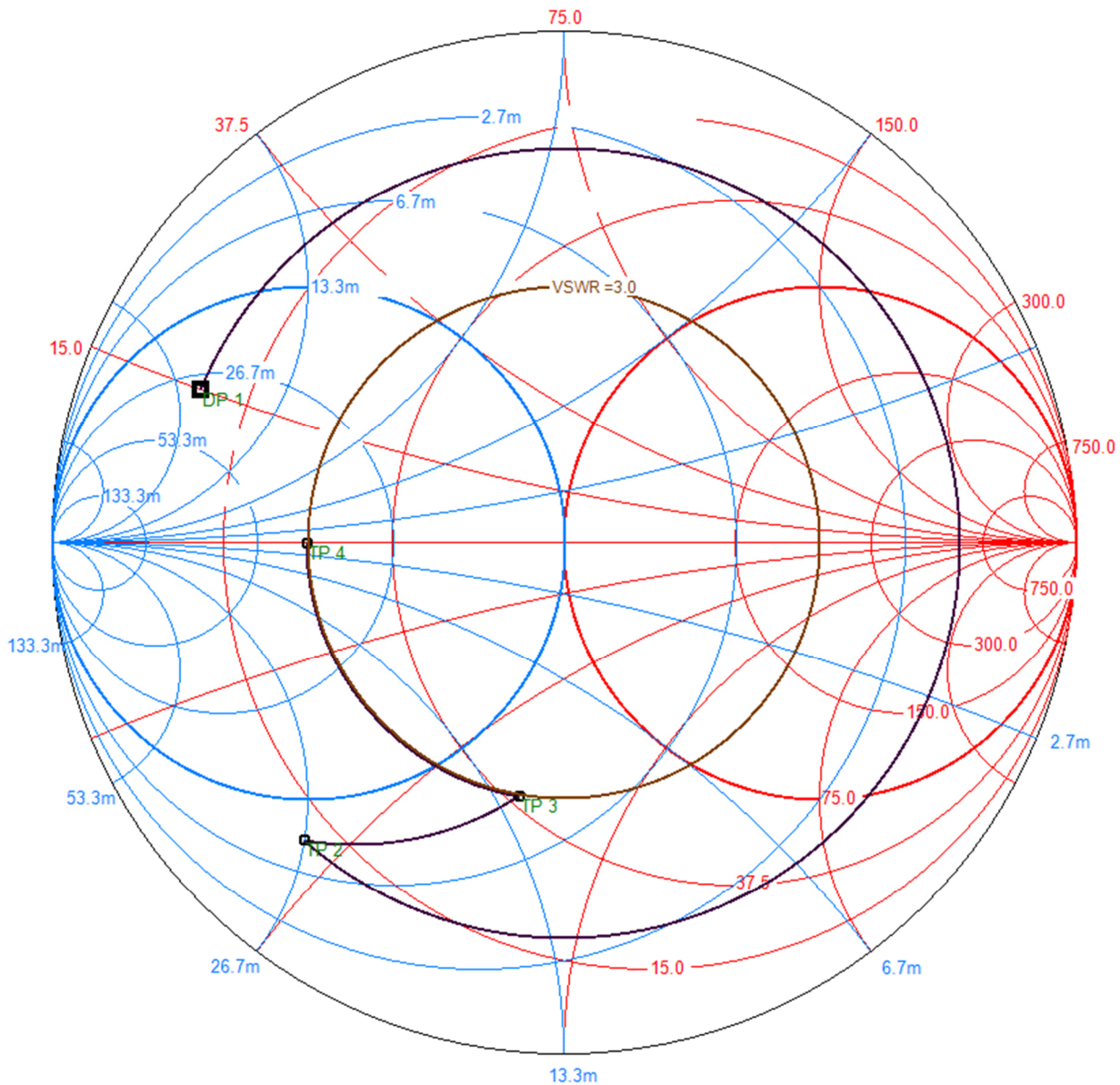


$$B_L \cdot Z_L = -1,401 \quad \rightarrow \quad |Z| = \omega L = \frac{Z_L}{1,401} \quad L = \frac{Z_L}{\omega \cdot 1,401} = \frac{75 \frac{\text{V}}{\text{A}}}{2\pi \cdot 500 \cdot 10^6 \frac{1}{\text{s}} \cdot 1,401} = \underline{\underline{17,0 \text{ nH}}}$$

$$\frac{X_C}{Z_L} = -0,305 \quad \rightarrow \quad |Y| = \omega C = \frac{1}{0,305 \cdot Z_L} \quad C = \frac{1}{\omega \cdot 0,305 \cdot Z_L} = \frac{1}{2\pi \cdot 500 \cdot 10^6 \frac{1}{\text{s}} \cdot 0,305 \cdot 75 \frac{\text{V}}{\text{A}}} = \underline{\underline{13,9 \text{ pF}}}$$

2. Punkt 1 mit L und l_2 im Smith-Chart in Z_e transformieren :



aus Smith-Chart abgelesen:

$$B'_L \cdot Z_L = -1,087 \quad \rightarrow \quad |Z| = \omega L' = \frac{Z_L}{1,087} \quad L' = \frac{Z_L}{\omega \cdot 1,087} = \frac{75 \frac{\text{V}}{\text{A}}}{2\pi \cdot 500 \cdot 10^6 \frac{1}{\text{s}} \cdot 1,087} = \underline{\underline{22,0 \text{ nH}}}$$

$$\frac{l'_2}{\lambda} = 0,1108 \quad \rightarrow \quad l'_2 = 0,1108 \cdot 400 \text{ mm} = \underline{\underline{44,3 \text{ mm}}}$$

Aufgabe 2:

(19 Punkte)

1. mit 3-6 für $Z_L = 75\Omega$: $\frac{w}{h} = 0,9 \rightarrow \underline{\underline{w}} = 0,9 \cdot h = 0,9 \cdot 0,8\text{mm} = \underline{\underline{0,72\text{mm}}}$

mit 3-5 für $\frac{w}{h} = 0,9$: $\varepsilon_{r,eff} = 3,2$

$$\underline{\underline{\lambda}} = \frac{c_0}{\sqrt{\varepsilon_{r,eff}} \cdot f} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{3,2} \cdot 3,0 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{56\text{mm}}}$$

2. $\frac{s}{h} = \frac{3,2\text{mm}}{0,8\text{mm}} = 4,0$ mit 7-14: $\underline{\underline{C_{min} = 30,2\text{dB}}}$

3. $C = 40\text{dB} \rightarrow |\underline{s}_{41}| = 10^{\frac{40\text{dB}}{20\text{dB}}} = 10^{-2} \quad k_L = 10^{\frac{30\text{dB}}{20\text{dB}}} = 3,16 \cdot 10^{-2}$

$$|\underline{s}_{41}|^2 = \frac{k_L^2 \cdot \sin^2 \beta l_m}{(1 - k_L^2) \cdot \cos^2 \beta l_m + \sin^2 \beta l_m}$$

für $k_L \ll 1$: $|\underline{s}_{41}|^2 = \frac{k_L^2 \cdot \sin^2 \beta l_m}{\cos^2 \beta l_m + \sin^2 \beta l_m} = \frac{k_L^2 \cdot \sin^2 \beta l_m}{1} = k_L^2 \cdot \sin^2 \beta l_m$

$$\sin^2 \beta l_m = \frac{|\underline{s}_{41}|^2}{k_L^2}; \quad \sin \beta l_m = \frac{|\underline{s}_{41}|}{k_L}; \quad \beta = \frac{2\pi}{\lambda}$$

$$l_m = \frac{1}{\beta} \cdot \arcsin\left(\frac{|\underline{s}_{41}|}{k_L}\right) = \frac{\lambda}{2\pi} \cdot \arcsin\left(\frac{|\underline{s}_{41}|}{k_L}\right) = \frac{56\text{mm}}{2\pi} \cdot \arcsin\left(\frac{10^{-2}}{3,16 \cdot 10^{-2}}\right) = 2,9\text{mm}$$

für $C > 40\text{dB}$ darf die Länge l also maximal um $\pm l_m$ von $k \cdot \lambda/2$ abweichen.

$$3 \cdot \frac{\lambda}{2} = 84\text{mm} < 100\text{mm} \quad 4 \cdot \frac{\lambda}{2} = \underline{\underline{112\text{mm}}} \quad 5 \cdot \frac{\lambda}{2} = 140\text{mm} > 130\text{mm}$$

$$\rightarrow 112\text{mm} - l_m < l < 112\text{mm} + l_m \quad \text{also: } \underline{\underline{109,1\text{mm} < l < 114,9\text{mm}}}$$

Aufgabe 3:

(21 Punkte)

1.

$$\underline{\underline{Z_E}} = Z_L \cdot \frac{1 + \underline{s_{11}}}{1 - \underline{s_{11}}} = 50\Omega \cdot \frac{1 + 0,54 \cdot e^{-j157^\circ}}{1 - 0,54 \cdot e^{-j157^\circ}} = (15,5 - j9,23)\Omega = \underline{\underline{18,0\Omega \cdot e^{-j30,8^\circ}}}$$

$$\underline{\underline{Z_A}} = Z_L \cdot \frac{1 + \underline{s_{22}}}{1 - \underline{s_{22}}} = 50\Omega \cdot \frac{1 + 0,38 \cdot e^{-j79^\circ}}{1 - 0,38 \cdot e^{-j79^\circ}} = (42,8 - j37,3)\Omega = \underline{\underline{56,8\Omega \cdot e^{-j41,1^\circ}}}$$

$$\underline{Z_E} = Z_{L,TV} \cdot \frac{1 + \underline{s_{11,TV}}}{1 - \underline{s_{11,TV}}}; \quad 1 + \underline{s_{11,TV}} = (1 - \underline{s_{11,TV}}) \cdot \frac{\underline{Z_E}}{Z_{L,TV}} = \frac{\underline{Z_E}}{Z_{L,TV}} - \frac{\underline{Z_E}}{Z_{L,TV}} \cdot \underline{s_{11,TV}}$$

$$\left(1 + \frac{\underline{Z_E}}{Z_{L,TV}}\right) \cdot \underline{s_{11,TV}} = \frac{\underline{Z_E}}{Z_{L,TV}} - 1; \quad \underline{s_{11,TV}} = \frac{\frac{\underline{Z_E}}{Z_{L,TV}} - 1}{\frac{\underline{Z_E}}{Z_{L,TV}} + 1} = \frac{\underline{Z_E} - Z_{L,TV}}{\underline{Z_E} + Z_{L,TV}}$$

$$\underline{\underline{\underline{s_{11,TV}}}} = \frac{\underline{Z_E} - Z_{L,TV}}{\underline{Z_E} + Z_{L,TV}} = \frac{(15,5 - j9,23)\Omega - 75\Omega}{(15,5 - j9,23)\Omega + 75\Omega} = \underline{\underline{0,66 \cdot e^{-j165^\circ}}} = (-0,64 - j0,17)\Omega$$

$$\text{analog: } \underline{\underline{\underline{s_{22,TV}}}} = \frac{\underline{Z_A} - Z_{L,TV}}{\underline{Z_A} + Z_{L,TV}} = \frac{(42,8 - j37,3)\Omega - 75\Omega}{(42,8 - j37,3)\Omega + 75\Omega} = \underline{\underline{0,40 \cdot e^{-j113^\circ}}} = (-0,16 - j0,37)\Omega$$

2.

$$\underline{\underline{V_{Tr,max}}} = |\underline{s_{21}}|^2 \cdot \frac{1}{1 - |\underline{s_{11}}|^2} \cdot \frac{1}{1 - |\underline{s_{22}}|^2} = 5,6^2 \cdot \frac{1}{1 - 0,54^2} \cdot \frac{1}{1 - 0,38^2} = 51,74 \quad [17,1\text{dB}]$$

$$\underline{\underline{V_{Tr,max}}} = |\underline{s_{21,TV}}|^2 \cdot \frac{1}{1 - |\underline{s_{11,TV}}|^2} \cdot \frac{1}{1 - |\underline{s_{22,TV}}|^2}$$

$$|\underline{s_{21,TV}}|^2 = V_{Tr,max} \cdot (1 - |\underline{s_{11,TV}}|^2) \cdot (1 - |\underline{s_{22,TV}}|^2)$$

$$\underline{\underline{|\underline{s_{21,TV}}|}} = \sqrt{V_{Tr,max} \cdot (1 - |\underline{s_{11,TV}}|^2) \cdot (1 - |\underline{s_{22,TV}}|^2)} = \sqrt{51,74 \cdot (1 - 0,70^2) \cdot (1 - 0,40^2)} = \underline{\underline{4,7}}$$

3.

$$\begin{aligned} |\underline{s_{11}}\underline{s_{22}} - \underline{s_{12}}\underline{s_{21}}|_{min} &= |0,70 \cdot 0,40 \cdot e^{j(\varphi_{11} + \varphi_{22})} - 0,1 \cdot 5,0 \cdot e^{j(\varphi_{12} + \varphi_{21})}| = \\ &= |(0,70 \cdot 0,40 - 0,1 \cdot 5,0) \cdot e^{j(\varphi_{12} + \varphi_{21})}| = \\ &= |(0,70 \cdot 0,40 - 0,1 \cdot 5,0)| = 0,22 \end{aligned}$$

$$K = \frac{1 - |\underline{s_{11}}|^2 - |\underline{s_{22}}|^2 + |\underline{s_{11}}\underline{s_{22}} - \underline{s_{12}}\underline{s_{21}}|_{min}}{2|\underline{s_{12}}\underline{s_{21}}|} = \frac{1 - 0,70^2 - 0,40^2 + 0,22^2}{2 \cdot 0,1 \cdot 5,0} = 0,3984 < 1$$

Transistor ist **nicht** absolut stabil!

Aufgabe 4:

(14 Punkte)

1.

$$\begin{aligned}
 F_{LNB} &= 10^{\frac{1,5\text{dB}}{10\text{dB}}} = 1,41 & G_{LNB} &= 10^{\frac{55\text{dB}}{10\text{dB}}} = 316 \cdot 10^3 \\
 \left(G_L = 10^{\frac{10\text{dB}}{10\text{dB}}} = 0,1 \right) & & F_L &= \frac{1}{G_L} = 10 \\
 \underline{\underline{F_{ges}}} &= 1 + (F_{LNB} - 1) + \frac{F_L - 1}{G_{LNB}} = 1 + (1,41 - 1) + \frac{10 - 1}{316 \cdot 10^3} = 1,41 \triangleq 1,5\text{dB}
 \end{aligned}$$

2.

$$\begin{aligned}
 P_{R,ein} &= k \cdot T_A \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 35\text{K} \cdot 27 \cdot 10^6 \frac{1}{\text{s}} = 1,304 \cdot 10^{-14} \text{W} \\
 \left(\frac{S}{N} \right)_{ein} &= F_{ges} \cdot \left(\frac{S}{N} \right)_{aus} = 1,5\text{dB} + 15\text{dB} = 16,5\text{dB} \\
 \rightarrow \underline{\underline{P_{S,ein}}} &= P_{R,ein} \cdot 10^{\frac{16,5\text{dB}}{10\text{dB}}} = 1,304 \cdot 10^{-14} \text{W} \cdot 10^{\frac{16,5\text{dB}}{10\text{dB}}} = 5,825 \cdot 10^{-13} \triangleq \underline{\underline{-92,3\text{dBm}}}
 \end{aligned}$$

3. Der Verstärker verstärkt Signal und Rauschen in gleichem Maß. Das S/N verbessert sich dadurch nicht (es verschlechtert sich aber auch nicht wesentlich).
 → es ist immer noch dieselbe Signalstärke von der Antenne erforderlich,
 also um 0dB geringer

Aufgabe 5:

(16 Punkte)

1. Reflexionsfaktor der Ersatzlast:

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = j0,7 + \frac{(-j0,3) \cdot (-j0,3) \cdot j0,6}{1 - j0,7 \cdot j0,6} = j0,7 + \frac{-j0,054}{1 + 0,42} = j0,6620$$

$$\underline{\underline{P'_Q}} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r}_1|^2}{|1 - \underline{r}_Q \cdot \underline{r}_1|^2} = 1,0 \text{ W} \cdot \frac{1 - 0,6620^2}{|1 - j0,5 \cdot j0,6620|^2} = \underline{\underline{317 \text{ mW}}}$$

2. Ersatzquelle:

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = j0,7 + \frac{(-j0,3) \cdot (-j0,3) \cdot j0,5}{1 - j0,7 \cdot j0,5} = j0,7 + \frac{-j0,045}{1 + 0,35} = j0,6667$$

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b}_Q \cdot \frac{-j0,3}{1 - j0,7 \cdot j0,5} = \underline{b}_Q \cdot (-j0,2222)$$

$$\underline{\underline{P'_L}} = \frac{1}{2} |\underline{b}_{Q2}|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_2 \cdot \underline{r}_L|^2} = 1,0 \text{ W} \cdot (-0,2222)^2 \cdot \frac{1 - 0,6^2}{|1 - j0,6667 \cdot j0,6|^2} = \underline{\underline{16,13 \text{ mW}}}$$

$$3. \quad \underline{\underline{a_B}} = 10 \text{ dB} \cdot \lg \left(\frac{P_{Q,abs}}{P_{L,abs}} \right) = 10 \text{ dB} \cdot \lg \left(\frac{P'_Q}{P'_L} \right) = 10 \text{ dB} \cdot \lg \left(\frac{300 \text{ mW}}{12,5 \text{ mW}} \right) = \underline{\underline{13,8 \text{ dB}}}$$

ohne 2-Tor:

$$P_{L,abs0} = \frac{1}{2} |\underline{b}_Q|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_Q \cdot \underline{r}_L|^2} = 1,0 \text{ W} \cdot \frac{1 - 0,6^2}{|1 - j0,5 \cdot j0,6|^2} = 379 \text{ mW}$$

$$\underline{\underline{a_E}} = 10 \text{ dB} \cdot \lg \left(\frac{P_{L,abs0}}{P_{L,abs}} \right) = 10 \text{ dB} \cdot \lg \left(\frac{P_{L,abs0}}{P'_L} \right) = 10 \text{ dB} \cdot \lg \left(\frac{379 \text{ mW}}{12,5 \text{ mW}} \right) = \underline{\underline{14,8 \text{ dB}}}$$